

MODEL QUESTION PAPER

REVISION. 2010
SUB CODE:3060

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY

PROGRAMMING IN C

(Common to AE, EA, EL, EC, EP, MD, TC)

Time : 3 hrs

Maximum marks : 100

PART A

(Answer all questions in one or two sentences. Each question carries 2 marks)

I)

- 1) Write the general format for scanf?
- 2) How is the execution of a while loop terminated?
- 3) Mention the use of ?: operator?
- 4) Illustrate the initialization of a two dimensional array?
- 5) Write the format for addressing a variable through its pointer?

(5 x 2 = 10)

PART B

(Answer any *five* of the following questions. Each question carries 6 marks)

II)

- 1) Identify the assignment, declaration statements and errors in the following program which calculate simple interest.

```
main ( );  
{  
int p,n,si;  
float r;  
P = 15,000;  
n = 5;  
r = 6.10;  
si = pnr/100;  
printf ( %f, si)  
}
```

- 2) Calculate the sum of every n^{th} integer beginning with n_{start} (ie, $i = n_{\text{start}}, n_{\text{start}}+n, n_{\text{start}} + 2* n, \dots$). Continue the looping process for all values of i that do not exceed n_{stop} ?
- 3) Evaluate the given expression using one dimensional array y to read the values and calculate the sum of their cubes.

$$\text{Total} = \sum_{i=1}^{20} y_i^3$$

- 4) Describe the format for declaring, opening and closing a file?
- 5) What is a union ? How does it differ from a structure?
- 6) Write a loop that will generate every third integer beginning with $i=2$, and continuing for all integer that are less than 100. Calculate the sum of those integers that are evenly divisible by i ? Use the conditional operator.
- 7) How are multidimensional arrays defined? Compare with the manner in which one dimensional array are defined?

(5x 6 = 30)

PART C

(Answer **one** full question from each module. Each question carries 15 marks)

MODULE I

- III)a) What is meant by operator precedence? List the operators according to their precedence? 10
b) Write a c program to find area of a sphere? Assume radius as 'a'? 5

OR

- IV)a) Explain four basic types of constants in C? 10
b) Write a C program to display the following output. 5
1234
234
34
4

MODULE II

- V) What are the conditional control transfer statements in C? When are they used? 15

OR

- VI)a) Write a loop that will calculate the sum of every third integer beginning with i=2.(ie, Calculate the sum 2+5+8+11+.....) for all values of i that are less than 100?Write the three different ways? 15

MODULE III

- VII)a)What is the difference between a string and an array of character/ write the syntax for both? 8
b) Write a program to concatenate two strings? 7

OR

- VIII)a) Consider the following program for comparing 2 strings. State whether the code will produce the expected output. If not identify the errors and correct it? 10

```
main ( )  
{  
char str1 [ ]= "Hello\r";  
char *str2 = Hello \t;  
printf(" d",strcmp{str1,str2}  
&&strcmp (str1,str2)  
}
```

- b) Write short note on arrays? 5

MODULE IV

- IX)a) Write a program to find the largest number using pointer? 9
b) Distinguish between the following functions

- i) getc and getchar
- ii) printf and fprintf
- iii) feof and ferror.

6

- X) a) Write a program to copy content of one file to another file? 15

OR

- b) Write the rules for initializing structures? 15